

ZACHARY TIANYI TANG

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EDUCATION

University of Toronto

B.S., Computer Science; GPA: 3.89/4.00

Toronto, ON

Expected May 2028

WORK EXPERIENCE

UofT Dynamic Optimization & Operations Management Lab

Researcher

Toronto, ON

Sep 2025 - Present

- Key member driving **robotics and Vision-Language-Action (VLA) industry research** aimed at automating large-scale **car battery assembly using UR5 robotic arm with Prof. Chi-Guhn Lee**
- Engineered simulation environment in **IsaacSim/Lab**, establishing a **teleoperation pipeline** that generated the **critical training data necessary** to successfully transfer robotic policies to **physical hardware**
- Specializing in fine-tuning open-source VLA models ($\pi_{0.5}$) to develop a generalizable wire manipulator

Philer AI

Software Engineer

Remote

Apr 2025 - June 2025

- Spearheaded a **10+ member team** to **successfully deploy an end-to-end customer intake legal AI assistant to production**, managing the entire stack including **full-stack deployment on Google Cloud/AWS Lambda**
- Executed **experimental RAG research** on advanced retrieval methodologies, including **multi-step retrieval** and **prompt decomposition** to facilitate **complex user legal queries over a large corpus of legal documents**
- Developed a custom, explainable RAG evaluation framework on **Pinecone**, benchmarking performance across metrics like correctness and recall, achieving an **average metric boost of 18%** through hyperparameter tuning
- Built frontend with **Next.js/React**, backend with **Python/Flask** including **LangGraph multi-LLM-agent workflow**

FORMS Syntion Information HK

MLE Intern

Hong Kong SAR, China

May 2025 - June 2025

- Developed internal tool for **HSBC's largest branch (US\$3T+ in assets, 20,000+ employees)** to automate extraction of key data from unstructured police fraud reports for **financial fraud detection**
- Engineered a **distributed fine-tuning pipeline** for 70B LLM models using **LoRa and Transformers** to automate fraud data extraction, deploying and managing **large-scale SLURM jobs across an A100 GPU cluster**
- Led **weekly client meetings** to align evaluation metrics with their criteria, and boosted model accuracy on those metrics from **~70% up to 97%** through **few-shot prompt engineering and hard negative mining**
- Built **synthetic data pipeline** constructed from proprietary police report segments, growing dataset from **250 to 10,000+ realistic data samples** to improve domain-specific LLM performance

PROJECTS & EXTRACURRICULARS

Heimer Academy | *React, Vite, FastAPI, Supabase, Amazon SageMaker, Amazon Bedrock* ([Devpost](#)) ([GitHub](#))

- 1st Place (\$10,000 USD + \$10k AWS Credits)** at **AWS x Riot Games Rift Rewind** global hackathon
- Engineered champion recommendation system using **co-occurrence graph analysis** across tens of thousands of player profiles and **pairwise ability similarity scoring** over **600+ abilities (200,000+ comparisons)**

Real-Time Video Super-Resolution | *PyTorch, TensorRT, CUDA, C++, OpenCV*

- Spearheading **real-time video super-resolution** research optimizing test-time inference with **TensorRT and CUDA**
- Profiled model bottlenecks using **NVIDIA Nsight Compute** and **PyTorch Profiler** to find high-latency operations
- Developed low-latency **C++/C** video capture pipeline with **OpenGL** to minimize **GPU/CPU overhead**

Multi-Agent RL Communication in Overcooked | *JAX, Python*

- Co-leading research in collaboration with **Prof. Matthew Taylor at the University of Alberta** introducing novel Overcooked benchmark focused on **multi-agent RL communication**
- People's Choice Award** at **CUCAI 2026**, Canada's largest undergrad AI conference (**340+ delegates, 19 schools**)

TECHNICAL SKILLS

Programming Languages: Python, JavaScript, Java, SQL, TypeScript, C#, C++

AI/ML Frameworks: PyTorch, Unsloth, Hugging Face, LangGraph, OpenCV, NumPy, SciPy, IsaacLab, IsaacSim, USDA

Full-stack Development: React, Next.js, Flask, Django, HTML/CSS, Tailwind CSS, shadcn/ui

Databases & Tools: SQLite, Git, Docker, SLURM, Pandas, jsPDF, Linux, Bash, Ubuntu, vim